

The Metrics:

ROC-AUC Score: 87.2% (Strong overall accuracy)

Recall (at 0.10 threshold): 92.9% (Caught almost all fraud)

Optimization: SMOTE + scale\_pos\_weight applied

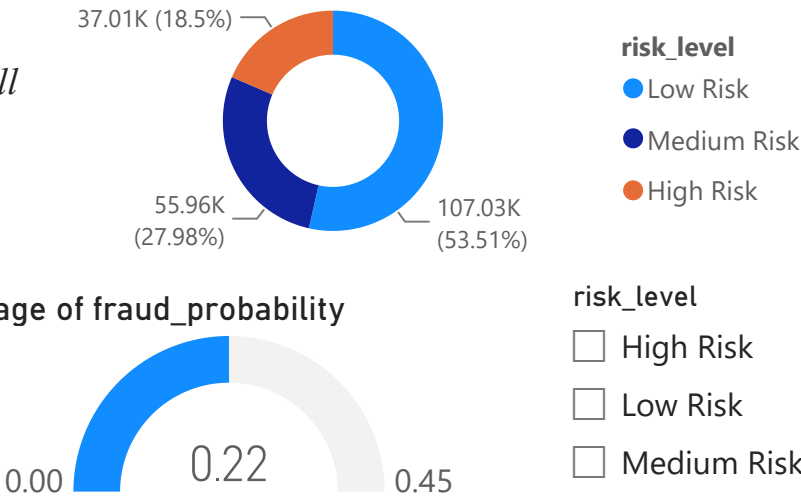
200K

Count of fraud\_...

93K

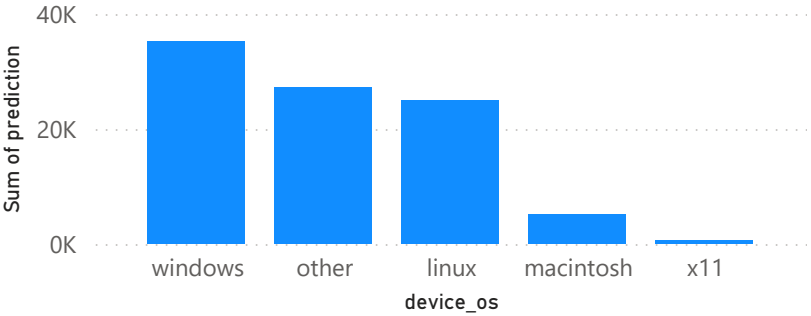
Sum of predic...

Count of fraud\_bool by risk\_level

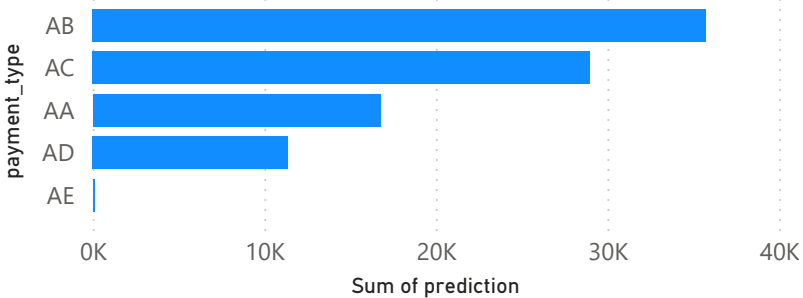


| device_os | payment_type | credit_risk_score | fraud_probability |
|-----------|--------------|-------------------|-------------------|
| windows   | AA           | 102               | 1.00              |
| windows   | AC           | 206               | 1.00              |
| windows   | AA           | 26                | 1.00              |
| other     | AB           | 102               | 1.00              |
| other     | AC           | 106               | 1.00              |
| windows   | AC           | 299               | 1.00              |
| windows   | AC           | 302               | 1.00              |
| windows   | AC           | 270               | 1.00              |
| windows   | AC           | 316               | 1.00              |
| windows   | AC           | 349               | 1.00              |
| windows   | AC           | 261               | 1.00              |
| windows   | AA           | 62                | 1.00              |
| windows   | AC           | 165               | 1.00              |

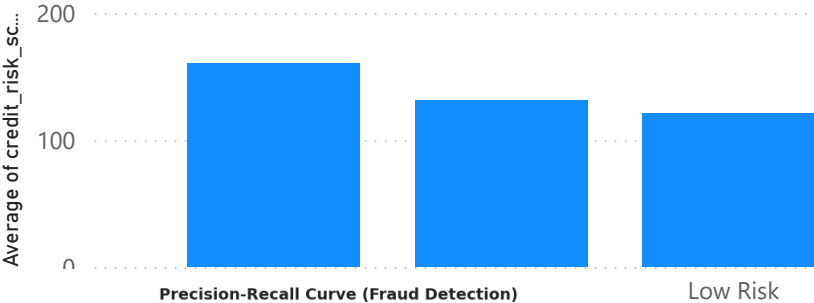
Sum of prediction by device\_os



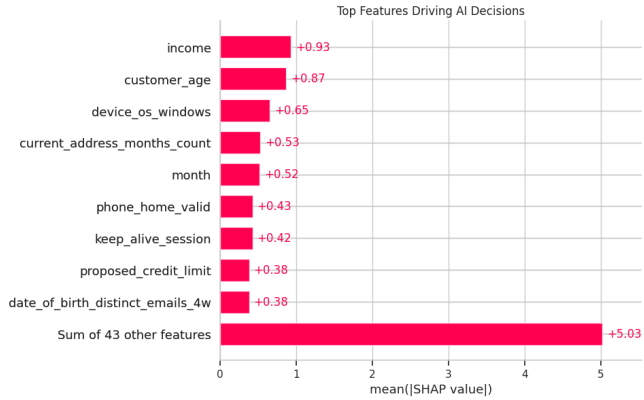
Sum of prediction by payment\_type



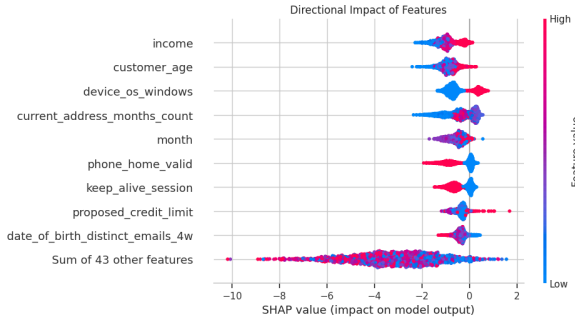
Average of credit\_risk\_score by risk\_level



1. GLOBAL: WHICH FEATURES INFLUENCE FRAUD THE MOST?



2. GLOBAL: HOW DO THEY INFLUENCE FRAUD?



Precision-Recall Curve (Fraud Detection)

